

Scalability & Future Plan

Date	8 July 2026
Team ID	XXXXXX
Project Name	Smart Lender
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Flask dev server	Single-threaded, not production-ready	High
2	No database	All predictions are lost after response	High
3	Static model.pkl	No automatic retraining on new data	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single Flask server, port 5000	Gunicorn + Cloud (Render/Railway/AWS)
2	Data Storage	No DB, stateless predictions	PostgreSQL + SQLAlchemy ORM
3	Performance	No caching or batch support	Redis caching + Celery async queues
4	Security	No auth, no HTTPS	Flask-Login + HTTPS + rate limiting

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	PostgreSQL database + prediction history page	1-2 months	Enables audit trail and persistent storage of all loan predictions
Phase 2	SHAP explainability per applicant	2-3 months	Improves transparency and trust in model decisions
Phase 3	User authentication + admin dashboard + bulk CSV upload	3-5 months	Supports multi-user access and batch processing for banks
Phase 4	Automated model retraining pipeline	6-9 months	Keeps model accuracy current with fresh loan data